

WI-FI USAGE ANALYZER

ABSTRACT

The **Wi-Fi Usage Analyzer** is a data analytics-based project designed to monitor, analyze, and visualize internet usage patterns in organizational networks such as colleges, hostels, and offices. The system uses network usage logs collected from managed Wi-Fi environments, containing information such as user identity, device details, MAC address, download and upload data, access time, and website or destination details. Using Python-based data analysis techniques with libraries including Pandas, NumPy, Matplotlib, and Seaborn, the project processes large volumes of network data to identify usage patterns such as top bandwidth consumers, peak usage hours, website category trends, and daily data consumption. The system also focuses on security by detecting abnormal network behaviour, including unusually high data uploads and large data transfers to unknown or unauthorized destinations, which may indicate possible malware activity, unauthorized access, or data leakage. An automated alert mechanism can notify administrators when suspicious activities or bandwidth limits are exceeded. The solution is designed to be scalable and can be extended for larger networks with more users and devices. Future enhancements include real-time monitoring, machine learning-based anomaly detection, and advanced threat analysis to improve network security and management.

Submitted By:

Team Members:

1. AKSHAY PUTTA

Roll No: 25B11CS022

2. J. DILEEP KUMAR

Roll No: 25B11CS342

3. Kancharla J S N S J Deepak

Roll No: 25B11CS390

4. P. KAMAL KIRAN

Roll No: 25B11CS744